

# ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

## TRAINING TR-102 REPORT DAY 9

5 August 2025

### Overview:

In today's session, we explored some important **statistical concepts** that are widely used in **data analysis and machine learning**. We learned how to calculate and understand **mean, median, mode**, and **standard deviation**—which help us summarize and understand datasets. We also learned about **percentiles** to see data distribution and practiced creating **scatter plots** to visualize relationships between variables. These concepts are essential for analyzing data before building machine learning models.

### Learning Objectives:

- Understand how to calculate and interpret mean, median, and mode
- Learn the purpose and calculation of standard deviation
- Use percentiles to analyze the spread and ranking in data
- Create and analyze a scatter plot to find patterns or correlations
- Strengthen foundational knowledge in data statistics for ML

### Mean (Average)

The mean is simply the average of all the numbers in a dataset.

We can find mean by adding up all the numbers and then divide by how many numbers there are.

Example:

If            the            numbers            are:            10,            20,            30

$$\text{Mean} = (10 + 20 + 30) \div 3 = 20$$

The mean helps us understand the overall value or central trend of the data.

## Median (Middle Value)

The median is the middle value in an ordered list of numbers.

- If the number of values is odd, it's the exact middle one.
- If it's even, you take the average of the two middle numbers.

Example:

List: 5, 10, 15 → Median = 10 (middle number)

List: 5, 10, 15, 20 → Median =  $(10 + 15) \div 2 = 12.5$

The median is useful when there are extreme values that might affect the mean.

## Mode (Most Frequent)

The mode is the number that appears most often in a dataset.

Example:

List: 1, 2, 2, 3, 4 → Mode = 2 (since it appears twice)

It's especially helpful in understanding what is common or frequent in the data.

## Standard Deviation

The standard deviation measures how spread out the numbers are from the mean.

- A small standard deviation means the values are close to the mean
- A large standard deviation means the values are spread out over a wide range

Real

Example:

If two classes have the same average marks (mean), but one class has students scoring 90 and 30, and the other has scores around 60, the first one has a higher standard deviation.

It helps understand data consistency or variation.

## Percentile (Relative Rank)

A percentile tells you how a particular value compares to the rest of the dataset. For example, being in the 90th percentile means you scored better than 90% of others.

Example:

If you're in the 75th percentile in an exam, your score is higher than 75% of the class.

It's useful in ranking, especially in exams, health stats, performance, etc.

## Scatter Plot (Data Visualization)

A scatter plot is a graph with dots that represent values for two different variables. It helps us see patterns, correlations, or outliers.

Example:

- X-axis: Study Time
- Y-axis: Exam Score  
→ The more time you study, the higher your marks → positive relationship

If the points go upward, it's a positive correlation.

If they go downward, it's a negative correlation.

If they're all scattered randomly, there may be no correlation.

## Conclusion:

Today's session was all about understanding how to make sense of data using basic statistical tools. These concepts may seem simple, but they play a big role in detecting trends, spotting errors, and preparing data for machine learning models. Now, terms like "standard deviation" or "percentile" won't feel confusing anymore! We also learn how powerful visualization can be, especially with scatter plots that give an instant view of relationships between variables. Overall, it was a very informative and hands-on day that built a strong base for future data projects.